

Ray Yan

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EDUCATION

UOW College Hong Kong

Associate of Science in Creative and Interactive Media Production

Hong Kong

2019 – 2021

SKILLS

Python Backend: Python (async/asyncio), FastAPI, Pydantic, pytest, Django, service-oriented code organization; Node.js, Express, Fastify, Java (J2EE)

LLM & Agents: Azure OpenAI / OpenAI APIs (chat completions, embeddings, streaming, tool calling, structured output, retries/timeout handling, token & cost tracking), LangChain, LangGraph, LlamaIndex, vLLM, RAG pipelines, MCP integration, Dify, Hugging Face, n8n

Observability & Monitoring: Langfuse (LLM tracing, prompt/version tracking, tool-call failures, token/cost dashboards), Grafana, Loki, Prometheus, Splunk, alerting

Data & SQL: PostgreSQL, Cosmos DB for PostgreSQL, MySQL, Databricks SQL Warehouse, Redis (caching), MongoDB, vector databases; SQL for business logic, validation, aggregation, and reusable templates

Search & Retrieval: Azure AI Search, vector search, keyword search, hybrid retrieval, embeddings, ranking, query performance optimization

Cloud & Azure: Azure (OpenAI, AI Search, Cosmos DB, Databricks, DevOps Pipelines, storage, service principals, managed identity), AWS (CLP, SAA certified), GCP, DEV/UAT/PROD environment setup

Security & Access Control: OAuth 2.0 / Entra ID, service principals, secure token handling, backend access validation, sensitive error masking, HashiCorp Vault, CyberArk

CI/CD & Infrastructure: GitHub Actions, Jenkins (Groovy), GitLab CI, Azure DevOps, Docker, Kubernetes (CKAD certified), Terraform, Helm, Ansible, SonarQube, JIRA

Testing: pytest, Selenium, Playwright; unit, integration, API, and stress testing; test automation in CI

Frontend: TypeScript, React, Next.js, Vue – for UI-backend integration and middle-layer work

Languages: Chinese (Cantonese): Native, Chinese (Mandarin): Native, English: Fluent

EXPERIENCE

Prudential (Hong Kong) Limited

Platform Engineer, Data Transformation Scrum Team

Hong Kong

Nov 2025 – Current

- Productionize LLM and agent workflows into reliable backend services and pipelines using async Python (FastAPI, Pydantic), collaborating with data scientists, LLM developers, frontend engineers, and QA in Agile sprints (JIRA) within a large-scale financial services enterprise.
- Integrate Azure OpenAI / OpenAI APIs into platform services – chat completions, embeddings, streaming, tool calling, and structured output – with retry/timeout handling, failsafe fallback behavior, and per-request token and cost tracking.
- Built Retrieval-Augmented Generation (RAG) pipelines with LangChain and vector databases for enterprise knowledge retrieval, covering embeddings, vector/keyword/hybrid retrieval, ranking, and query performance optimization; self-hosted open-source LLMs with vLLM (CUDA/GPU-accelerated) for high-throughput, low-latency inference.
- Supported the MLOps team in deploying and configuring Azure AI Search and Cosmos DB for PostgreSQL as retrieval and storage backends for AI services, including index/schema setup and environment provisioning across DEV/UAT/PROD.
- Set up and manage an AI Gateway middle layer (Kong and Bifrost) between UI clients and agent/model backends for centralized LLM routing, session and authentication handling, rate limiting, request validation, and service-to-service communication; implemented Guardrails for prompt/response validation, PII redaction, and sensitive error masking.
- Develop and manage data pipelines and analytics workloads on Azure Databricks (Spark, Databricks SQL), writing SQL for business logic, validation, aggregation, and reusable templates that integrate AI/ML model outputs into business use cases such as marketing analytics.

- Monitor LLM and platform health with Grafana dashboards (Loki, Prometheus) covering latency, error rates, tool-call failures, and token/cost consumption, with alerting; identify and resolve performance bottlenecks via caching (Redis), async processing, and infrastructure tuning.
- Operate CI/CD pipelines (GitHub Actions, Jenkins) with Terraform and Helm for Kubernetes, automating build, test, security scanning, and release/version-tracked deployments of LLM serving infrastructure and gateway components across DEV/UAT/PROD environments.
- Conduct proof-of-concept evaluations of emerging AI tooling and contribute to planning, documentation, and PR reviews to improve platform scalability, reliability, and developer experience.

PCCW Solutions

Hong Kong

DevOps Engineer

Mar 2025 – Nov 2025

- DevOps Engineer for the largest Hong Kong eMPF Platform – a central electronic platform for MPF scheme administration in a large-scale financial services enterprise environment.
- Designed and operated CI/CD pipelines using Jenkins and Git (Bitbucket) with unit testing, integration testing, artefact management, SonarQube security scanning, and release-tracked deployments across DEV/UAT/PROD environments; provided production incident support and root-cause analysis.
- Created automation and test scripts using Python, Bash, and Jenkins Groovy while implementing Docker containerization. Managed deployment lifecycle on Red Hat OpenShift (Kubernetes) with Oracle Database environments.
- Integrated DevSecOps tooling including CyberArk and HashiCorp Vault for secure secret and token handling, SonarQube for code quality, and Splunk for security monitoring throughout CI/CD pipelines. Coordinated workflows using JIRA.
- Collaborated with cross-functional Application Teams, Architects, QA, and vendor teams on version control, automated testing, and environment setup in a continuous delivery environment, with strong documentation and PR review discipline.
- Deployed and configured web applications on Nginx, MuleSoft API Gateway, and Solace. Managed service-to-service integrations with internal and external APIs within private cloud infrastructure.
- Monitored platform performance using Grafana and Loki dashboards with alerting, identifying and resolving latency and reliability bottlenecks to maintain high availability under concurrent user workloads.

Ntuple Global Limited

Hong Kong

Full-stack Developer/Senior Software Developer (Remote from Canada)

Oct 2022 – Mar 2025

- Designed and developed full-stack web applications (React/Next.js frontend, Node.js/Python backend, MySQL/PostgreSQL databases) for multiple F&B, CRM, and enterprise clients, optimizing API latency, database queries, and caching for high performance, scalability, and security.
- Implemented system architecture to extract terabytes of data for ETL pipeline development for the largest public railway transport in Hong Kong, writing SQL for validation, aggregation, and filtering and integrating with multiple data warehouse and service APIs.
- Developed async Python API services with automated testing (pytest, Selenium, Playwright) covering unit, integration, and API test scenarios; managed data storage in MySQL and Amazon S3.
- Managed AWS EKS (Kubernetes), EC2, and cloud infrastructure. Designed CI/CD pipelines using GitHub Actions with automated testing and AWS SSM integration. Utilized Terraform and Ansible for Infrastructure as Code.
- Developed online ordering systems with payment gateway API integrations (Mastercard, Stripe, GlobalPayment) following modern security standards.
- Led and mentored development teams of 2-3 people, participating in project planning sessions to analyse requirements, provide time estimates, and ensure code quality across multiple client projects.

PCCW Solutions Limited

Hong Kong

Software Engineer

Oct 2021 – Oct 2022

- Provided on-site production and User Acceptance Testing (UAT) support for Oracle Weblogic Server to maintain server stability and reliability.
- Maintained current Java codebase and fulfilled client requests.
- Developed a backend program with Java to generate CSV files from the database for security analysis.

Chun Wo Limited

Hong Kong

Web Developer

Jul 2021 – Oct 2021

- Designed and created a full-stack web application with PHP framework Laravel, along with React.js and Tailwind as the frontend.
- Performed basic network security testing to find vulnerabilities and patch before production.

CERTIFICATIONS

AWS Certified Solutions Architect – Associate: SAA - Expiration: 2027-07-20 - ID:
6571bd4dcd3542c084961a64105304f1

Linux Foundation Certified Kubernetes Application Developer: CKAD - ID: LF-sxumuxu0ty